

DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 6 (AT3)

Case Study

Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

Case Study 1 – Retail Sales Performance Dashboard

Scenario

A retail company has collected one year of sales data from five regions (North, South, East, West, Central). The dataset contains monthly sales, product categories, customer segments, and total revenue.

The management wants to identify:

- The highest-performing region.
- Monthly sales trends.
- Revenue contribution of each product category.
- Relationship between advertising expenditure and sales.
- Sales distribution across regions.

Tasks

1. Select the most suitable visualization for each requirement:
 - Regional revenue comparison
 - Monthly sales trend
 - Product category contribution
 - Advertising expenditure vs sales
 - Regional sales distribution
 2. Using **R**, create:
 - Bar Plot
 - Stacked Bar Chart
 - Histogram
 - Density Plot
 - Scatter Plot
 3. Interpret:
 - Which region generated the highest revenue?
 - Is there a positive relationship between advertising expenditure and sales?
 - Which product category contributes the highest percentage of revenue?
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Case Study 2 – University Student Performance Analysis

Scenario

A university stores examination data for four departments:

- Computer Science
- Information Technology
- Electronics
- Mechanical

The dataset includes:

- Semester marks
- Attendance percentage
- CGPA
- Placement status

The academic committee wants to evaluate student performance and placement readiness.

Tasks

1. Choose appropriate visualizations for:
 - Department-wise average marks

- Distribution of CGPA
 - Placement percentage
 - Attendance vs CGPA relationship
 - Semester-wise performance trend
2. Develop the following in **R**:
- Grouped Bar Chart
 - Violin Plot
 - Histogram with Density Curve
 - Scatter Plot
 - Line Chart
3. Interpret:
- Which department has the highest average CGPA?
 - Does higher attendance improve CGPA?
 - Which department has the highest placement rate?
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Case Study 3 – Hospital Patient Analytics

Scenario

A multi-specialty hospital records patient information including:

- Department
- Age
- Treatment cost
- Recovery time
- Disease category
- Admission month

Hospital administrators want to improve resource planning.

Tasks

1. Select suitable visualizations for:
- Monthly patient admissions
 - Distribution of treatment costs
 - Disease category proportions
 - Age vs recovery time
 - Department-wise treatment costs
2. Create in **R**:
- Heatmap
 - Histogram
 - Violin Plot
 - Density Plot
 - Scatter Plot
3. Interpret:
- Which department has the highest treatment cost?
 - Which disease category occupies the largest proportion?
 - Is patient age associated with recovery time?
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Case Study 4 – Smart City Traffic Monitoring

Scenario

A smart city collects traffic sensor data from different junctions. The dataset contains:

- Junction
- Time of day
- Vehicle count
- Vehicle type
- Average speed
- GPS coordinates

City planners aim to reduce traffic congestion.

Tasks

1. Select suitable visualizations for:
 - Vehicle count comparison
 - Hourly traffic trend
 - Vehicle type proportion
 - Traffic density by location
 - Relationship between traffic volume and average speed
 2. Using **R**, create:
 - Bar Plot
 - Line Chart
 - Pie Chart
 - Geospatial Map
 - Scatter Plot
 3. Interpret:
 - Which junction experiences the highest congestion?
 - During which hours is traffic highest?
 - Does increased traffic reduce vehicle speed?
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Case Study 5 – Social Media User Interaction Analysis

Scenario

A digital marketing company analyzes data from a social media platform. The dataset includes:

- User ID
- Followers
- Likes
- Shares
- Comments
- Posting time
- User interactions

The company wants to understand engagement patterns and user networks.

Tasks

1. Select appropriate visualizations for:
 - Distribution of likes
 - Relationship between followers and engagement
 - User interaction network
 - Posting activity over time
 - Multivariate comparison of likes, shares, and comments

2. Develop in **R**:

- Histogram with Density Curve
- Scatter Plot
- Network Visualization
- Temporal Line Plot
- Heatmap

3. Interpret:

- Which users are the most influential?
- At what time is engagement highest?
- Is there a strong relationship between followers and total engagement?
- Identify highly connected users from the network visualization.
- Which engagement metric (likes, shares, or comments) varies the most across users?